



Worn Smooth, Rarely Gripped

A Time-Lapse Validation of the Running Path's Outdoor Fitness-Station Wear Index Against Observed Exercise Use

Kwame Lindqvist, Fenna Okoro

School of Emergent Priorities, Slop University

doi:10.5555/slop.r6jljv

Abstract. Facilities scores each of the University's 22 outdoor fitness stations on a fortnightly Wear Index — grip-tape abrasion, powder-coat paint loss, and ground compaction beneath the footplate — and treats a station's index as the closest available reading of how much it gets used, feeding a Horizon Register item that nominates sites for the running path's next installation wave ahead of any usage log Facilities has budgeted to collect. We test whether the Wear Index tracks exercise use at all, setting fourteen weeks of time-lapse imagery at all 22 stations against the index, with two blind coders separating each recorded interaction into a completed exercise set or a perch event — sitting, leaning, or waiting at the apparatus without exercising. The Wear Index correlates only weakly with exercise-set counts ($r = 0.24$, $n = 22$) and strongly with perch-event counts ($r = 0.77$), and the two highest-wear stations, both nominated in the live Horizon Register item, rank second and fifth of 22 for perch events against sixteenth and seventh for exercise sets. Recomputing the index from grip-tape abrasion alone, dropping the paint and compaction components, changes neither correlation by a distinguishable margin. We discuss what a wear signal that tracks contact rather than exercise may mean for a capacity-planning process built to anticipate demand ahead of a usage log.

Introduction

Facilities installed the University's first outdoor fitness stations — pull-up bars, dip stations, sit-up benches — along the running path in three waves since 2019, and now maintains 22 of them at uneven intervals around the 4.2-kilometre loop. No wave came with a budgeted usage log: a station's popularity was never something Facilities could query directly, only infer. In its place, a fortnightly Wear Index — three condition scores combined into one number — has become the closest reading Facilities has of how hard a station works, and the reading a live Horizon Register item now cites to nominate where the path's next stations should go, ahead of any firmer evidence a usage log might eventually supply.

The reasoning is not unreasonable on its face: a station gripped, leaned on, and stood beside more often should, all else equal, wear faster

than one nobody visits. What the reasoning elides is that grip-tape abrasion and ground compaction are indifferent to what generates the contact. A runner waiting for a group to assemble, resting a hand on the bar to stretch a calf, wears the same handle as a runner completing a set of pull-ups; a jacket dropped at the footplate compacts the same ground as a pair of boots planted for a dip. Wear reads as usage only if exercise is the dominant source of contact at a station, an assumption the Horizon Register item has not, to our knowledge, tested against anything.

We test whether the Wear Index agrees with actual exercise use, and, if it disagrees, what it is tracking instead.

We make three contributions, each hedged to what fourteen weeks at one path can show. First, we assemble a station-matched dataset pairing each station's Wear Index against inde-

pendently coded exercise-set and perch-event counts from a term of time-lapse imagery. Second, we show the index tracks perch events far more closely than it tracks exercise, and that this pattern holds for the two stations a live capacity-planning decision has already acted on. Third, we test whether the index's non-abrasion components are responsible for the perch association, and discuss what a wear signal built for maintenance scheduling may be doing when it is repurposed to anticipate demand.

Related work

Direct observation of park and recreation settings is well established, most prominently through SOPARC, whose reliability against simultaneous independent raters is high for participant counts and moderate-to-good for activity-level coding [1], and which has since anchored a large literature relating park characteristics to observed use [2]. The companion Environmental Assessment of Public Recreation Spaces instrument extends direct observation to a site's physical features, introduced with high inter-rater reliability [3] and later shown, in shortened form, to retain the original's correlation with observed physical activity [4], alongside a comparative audit of five such tools finding moderate-to-good inter-rater agreement across the family [5].

Outdoor fitness equipment specifically has a mixed usage record: in one direct-observation study, only 2.7% of a park's adult visitors were seen using installed fitness equipment across 100 observation-hours [6], and a substantial share of the interactions that are recorded do not match a station's intended exercise motion [7]. Natural experiments installing equipment find inconsistent downstream gains: visitation and activity increases are often statistically indistinguishable from a matched control [8], [9], though a diverse, age-targeted facility mix has shown clearer gains elsewhere [10] — even as older users report valuing the equipment for reasons a usage count alone would not surface [11].

Wear as a usage proxy has independent standing outside recreation-equipment research: protected-area managers have long read trail erosion as a proxy for hiking intensity, using exposed-root and substrate-loss measures precisely because a direct count is rarely available

[12], [13], the same trade-off the Wear Index makes on synthetic ground. Direct observation itself is not free of its own distortions — researcher presence measurably changes the behaviour under study [14] — and a measure adopted because it is available rather than because it is accurate is a recognised failure mode in institutional metrics generally [15]. Prior work at this institution has found two other physical-count proxies — a water-bubbler refill counter and a smoking shelter's monthly butt tally — correlate only weakly with the independently audited condition or behaviour they were meant to track [16], [17]; we extend this line to a multi-component condition rubric rather than an automated tap count.

Method

The study site is the University's running path, a 4.2-kilometre loop carrying 22 outdoor fitness stations installed in three waves since 2019: eleven pull-up-and-dip combination frames, six sit-up benches, and five balance-and-stretch units, spaced unevenly around the loop rather than at fixed intervals.

Facilities' Wear Index combines three condition scores, each rated 0-10 by two Facilities grounds staff working independently every second week across the fourteen-week study term (a full teaching term): grip-tape and handle-surface abrasion (G), powder-coat paint loss (P), and ground compaction beneath the station's foot-plate or landing zone (C), combined as

$$W_i = 0.4G_i + 0.3P_i + 0.3C_i,$$

and averaged across the term's seven rating rounds to give each station i a single mean Wear Index. Inter-rater agreement across all three components was high (Cohen's $\kappa = 0.86$).

Component	Weight	What it scores
Grip-tape abrasion (<i>G</i>)	0.4	Handle and bar surface wear from contact
Paint loss (<i>P</i>)	0.3	Powder-coat wear from weather and contact exposure
Ground compaction (<i>C</i>)	0.3	Soil compaction beneath the foot-plate or landing zone

Table 1: The Wear Index's three rated components and their combining weights.

Time-lapse and camera-trap imagery are established substitutes for continuous human observation in outdoor-recreation monitoring, validated against paired manual counts and video review [18], extended more recently to computer-vision-assisted camera-trap counting [19] and to crowd-sourced imagery for park-visitation estimation at a much larger scale [20]; we adopt the same logic at station scale. A time-lapse camera at each of the 22 stations captured one frame every two minutes from 6am to 8pm across the same fourteen weeks. Two coders, blind to every station's Wear Index and to each other's codes, reviewed each station's frame sequence and classified every distinct interaction as either an *exercise set* — a completed repetition sequence engaging the apparatus as designed — or a *perch event*: sitting, leaning, standing close, or otherwise occupying the apparatus without exercising, including waiting for a running group to assemble, adjusting footwear, or checking a phone. Coding disagreements (7.4% of interactions) were resolved by a third, independent review. Across the term the cameras yielded 833 coded exercise sets and 1,234 coded perch events over 2,067 total interactions.

We report the Pearson correlation between each station's mean Wear Index and its term-total exercise-set count, and separately against its term-total perch-event count. As an ablation on which Wear Index component is driving any association, we recompute the index from grip-tape abrasion alone (*G*, dropping *P* and *C*) and repeat both correlations, testing whether the

paint- and compaction-weighted components — both plausibly more sensitive to standing or sitting contact than to gripping and pulling — are responsible for any perch-event association the full index shows. Facilities' current Horizon Register item ranks all 22 stations by mean Wear Index alone and nominates the path's two highest-wear stations for the next installation wave; we report where those two stations fall on each of our independently coded measures.

Results

Across the fourteen-week term, station mean Wear Index ranged from 2.7 to 9.5 (mean 6.0, SD 1.8). Wear Index correlates only weakly with a station's total exercise-set count ($r = 0.24$, $n = 22$; Figure 1), a relationship a linear fit barely distinguishes from flat. The same index correlates strongly with total perch-event count ($r = 0.77$; Figure 2) — more than three times the exercise correlation's strength on the same 22 stations, over the same term, from the same rated instrument.

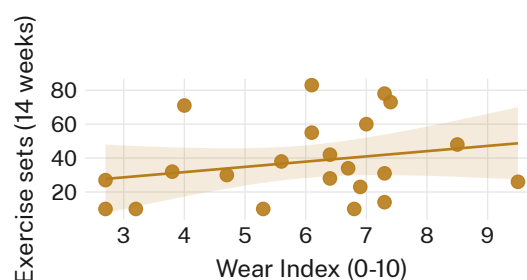


Figure 1: Station mean Wear Index against coded exercise-set count across the 14-week term ($n = 22$): the fitted line is barely distinguishable from flat ($r = 0.24$).

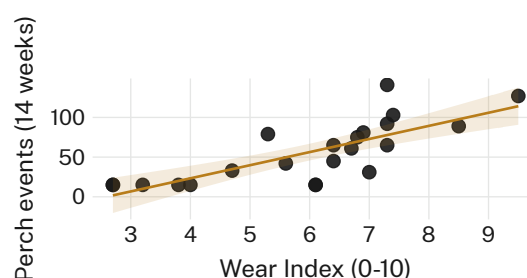


Figure 2: The same 22 stations' Wear Index against coded perch-event count: a far steeper, tighter relationship ($r = 0.77$).

The pattern holds at the two stations a live decision has already acted on. Facilities' Horizon Register item, filed nine weeks into the study term, ranks all 22 stations by mean Wear Index

and nominates the two highest — a pull-up-and-dip frame near the boathouse jetty ($W = 9.5$) and a sit-up bench near the oval car park ($W = 8.5$) — for the running path's next installation wave, citing each station's wear reading as evidence of unmet demand at its location. On our independently coded measures, the jetty station ranks second of 22 for perch events but sixteenth for exercise sets; the car-park station ranks fifth for perch events and seventh for exercise sets. Both sit within eighty metres of one of the path's two busiest group-run assembly points, where runners routinely arrive early and wait.

Recomputing the Wear Index from grip-tape abrasion alone, dropping the paint-loss and ground-compaction components, changes the picture by less than either correlation's own term-to-term noise would: the grip-tape-only index correlates with exercise sets at $r = 0.21$ (against the full index's $r = 0.24$) and with perch events at $r = 0.75$ (against $r = 0.77$; Figure 3). Removing two-thirds of the index's rated weight leaves its diagnostic character — weakly tracking exercise, strongly tracking perch use — statistically indistinguishable from the version Facilities currently rates; we retain the full three-component rubric for completeness, since grip-tape condition alone is no better a measure of exercise use than the index Facilities already runs.

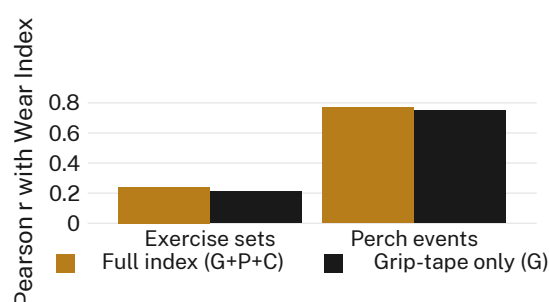


Figure 3: Recomputing the Wear Index from grip-tape abrasion alone barely moves either correlation: dropping two of three rated components changes the exercise correlation by 0.03 and the perch correlation by 0.02.

Discussion

These results suggest the Wear Index measures contact, not exercise, and that the two are only loosely related at the stations we studied. A runner who leans on a bar to stretch, or a group that gathers around a frame while waiting for stragglers, abrades grip tape and compacts

ground much as a set of pull-ups would, and plausibly for longer per visit; a wear signal built to flag maintenance need cannot, on this evidence, distinguish the two sources of contact from each other, which may be sufficient for its original purpose while being insufficient for the capacity-planning use the Horizon Register item has put it to. That the two stations already nominated for expansion rank near the top for perch events and only mid-table for exercise raises the possibility that the running path's next installation wave, as currently planned, would add capacity closer to where runners wait than to where they train. We do not read these results as evidence the Wear Index should be discontinued as a maintenance-scheduling instrument, a task it was built for and continues, on this evidence, to serve; the concern is narrower, and specific to its second, unaudited use as a demand signal.

Limitations

Our sample of 22 stations across a single running path and a single term is modest, though the exercise/perch asymmetry holds across all three station types (frames, benches, balance units) and both halves of the term, which we take as evidence against a one-off seasonal or cohort artefact. Time-lapse coding at two-minute intervals may undercount brief interactions shorter than the sampling gap; if anything this would understate rather than inflate both correlations equally, leaving their relative gap intact. Our coders, though blind to Wear Index scores, worked from the same footage Facilities' own raters did not see, so we cannot rule out that a differently instrumented rating process would separate perch and exercise contact more cleanly than the Index as currently specified.

Conclusion

We set 22 outdoor fitness stations' Facilities-rated Wear Index against fourteen weeks of independently coded exercise-set and perch-event counts, finding the index tracks perch use far more closely than exercise use, a pattern unchanged when the index is stripped to grip-tape abrasion alone. The two stations a live Horizon Register item has already nominated for the path's next installation wave rank high for perch events and only middling for exercise, suggesting the nomination may be reading rest-point demand as exercise demand. We suggest Facil-

ities decouple the Wear Index's maintenance-scheduling use, which this evidence does not challenge, from its capacity-planning use, which it does. Future work should test whether the same asymmetry holds on the University's other paths, and whether a dedicated exercise count materially changes the next wave's siting.

References

- [1] T. L. McKenzie, D. A. Cohen, A. Sehgal, S. Williamson, and D. Golinelli, "System for Observing Play and Recreation in Communities (SOPARC): Reliability and Feasibility Measures," *Journal of Physical Activity and Health*, vol. 3, no. s1, pp. S208–S222, 2006, doi: [10.1123/jpah.3.s1.s208](https://doi.org/10.1123/jpah.3.s1.s208).
- [2] K. R. Evenson, S. A. Jones, K. M. Holliday, D. A. Cohen, and T. L. McKenzie, "Park characteristics, use, and physical activity: A review of studies using SOPARC (System for Observing Play and Recreation in Communities)," *Preventive Medicine*, vol. 86, pp. 153–166, 2016, doi: [10.1016/j.ypmed.2016.02.029](https://doi.org/10.1016/j.ypmed.2016.02.029).
- [3] B. E. Saelens, L. D. Frank, C. Auffrey, R. C. Whitaker, H. L. Burdette, and N. Colabianchi, "Measuring Physical Environments of Parks and Playgrounds: EAPRS Instrument Development and Inter-Rater Reliability," *Journal of Physical Activity and Health*, vol. 3, no. s1, pp. S190–S207, 2006, doi: [10.1123/jpah.3.s1.s190](https://doi.org/10.1123/jpah.3.s1.s190).
- [4] C. M. Geremia, K. L. Cain, T. L. Conway, J. F. Sallis, and B. E. Saelens, "Validating and Shortening the Environmental Assessment of Public Recreation Spaces Observational Measure," *Journal of Physical Activity and Health*, vol. 16, no. 1, pp. 68–75, 2019, doi: [10.1123/jpah.2018-0142](https://doi.org/10.1123/jpah.2018-0142).
- [5] R. P. Joseph and J. E. Maddock, "Comparative Analysis of Five Observational Audit Tools to Assess the Physical Environment of Parks for Physical Activity, 2016," *Preventing Chronic Disease*, vol. 13, p. E169, 2016, doi: [10.5888/pcd13.160176](https://doi.org/10.5888/pcd13.160176).
- [6] J. L. Copeland, C. Currie, A. Walker, E. Mason, T. N. Willoughby, and A. Amson, "Fitness Equipment in Public Parks: Frequency of Use and Community Perceptions in a Small Urban Centre," *Journal of Physical Activity and Health*, vol. 14, no. 5, pp. 344–352, 2017, doi: [10.1123/jpah.2016-0277](https://doi.org/10.1123/jpah.2016-0277).
- [7] H.-wen Chow and D.-R. Wu, "Outdoor Fitness Equipment Usage Behaviors in Natural Settings," *International Journal of Environmental Research and Public Health*, vol. 16, no. 3, p. 391, 2019, doi: [10.3390/ijerph16030391](https://doi.org/10.3390/ijerph16030391).
- [8] J. Veitch, J. Salmon, G. Abbott, A. Timperio, and S. Sahlqvist, "Understanding the impact of the installation of outdoor fitness equipment and a multi-sports court on park visitation and park-based physical activity: A natural experiment," *Health & Place*, vol. 71, p. 102662, 2021, doi: [10.1016/j.healthplace.2021.102662](https://doi.org/10.1016/j.healthplace.2021.102662).
- [9] D. A. Cohen, T. Marsh, S. Williamson, D. Golinelli, and T. L. McKenzie, "Impact and cost-effectiveness of family Fitness Zones: A natural experiment in urban public parks," *Health & Place*, vol. 18, no. 1, pp. 39–45, 2012, doi: [10.1016/j.healthplace.2011.09.008](https://doi.org/10.1016/j.healthplace.2011.09.008).
- [10] C. Paudel, A. Timperio, J. Salmon, V. Loh, B. Deforche, and J. Veitch, "Something for the young and old: A natural experiment to evaluate the impact of park improvements," *Urban Forestry & Urban Greening*, vol. 100, p. 128486, 2024, doi: [10.1016/j.ufug.2024.128486](https://doi.org/10.1016/j.ufug.2024.128486).
- [11] H.-wen Chow, "Outdoor fitness equipment in parks: a qualitative study from older adults' perceptions," *BMC Public Health*, vol. 13, no. 1, p. 1216, 2013, doi: [10.1186/1471-2458-13-1216](https://doi.org/10.1186/1471-2458-13-1216).
- [12] M. Jula and M. Voiculescu, "Assessment of the Annual Erosion Rate along Three Hiking Trails in the Făgăraș Mountains, Romanian Carpathians, Using Dendrogeomorphological Approaches of Exposed Roots," *Forests*, vol. 13, no. 12, p. 1993, 2022, doi: [10.3390/f13121993](https://doi.org/10.3390/f13121993).
- [13] M. Jula and M. Voiculescu, "Assessment of the mean erosion rate using dendrogeomorphological approaches on exposed roots along hiking and biking trails in the Bucegi Mountains, Romanian Carpathians," *CATENA*, vol. 217, p. 106435, 2022, doi: [10.1016/j.catena.2022.106435](https://doi.org/10.1016/j.catena.2022.106435).

- [14] L. M. König, K. Pasko, K. Baga, R. Har-sora, and D. Arigo, "Isolating the role of researcher observation on reactivity to the measurement of physical activity," *Applied Psychology: Health and Well-Being*, vol. 17, no. 1, 2025, doi: [10.1111/aphw.12630](https://doi.org/10.1111/aphw.12630).
- [15] M. Fire and C. Guestrin, "Over-optimization of academic publishing metrics: observing Goodhart's Law in action," *GigaScience*, vol. 8, no. 6, 2019, doi: [10.1093/gigascience/giz053](https://doi.org/10.1093/gigascience/giz053).
- [16] M. Hanke and B. Ntuli, "Tapped Hardest, Fixed First: Auditing Whether Refill-Counter Reads Predict a Water-Fountain's Repair Priority," *Slop University technical report*, 2026, doi: [10.5555/slop.ajnbef](https://doi.org/10.5555/slop.ajnbef).
- [17] S. Okwuosa and T. Solberg, "One Bin, Fewer Butts: Auditing a Smoking Shelter's Falling Butt Count Against an Independent Campus-Wide Sweep," *Slop University technical report*, 2026, doi: [10.5555/slop.9a6na6](https://doi.org/10.5555/slop.9a6na6).
- [18] A. Arnberger, W. Haider, and C. Brandenburg, "Evaluating Visitor-Monitoring Techniques: A Comparison of Counting and Video Observation Data," *Environmental Management*, vol. 36, no. 2, pp. 317–327, 2005, doi: [10.1007/s00267-004-8201-6](https://doi.org/10.1007/s00267-004-8201-6).
- [19] J. Staab, E. Udas, M. Mayer, H. Taubenböck, and H. Job, "Comparing established visitor monitoring approaches with triggered trail camera images and machine learning based computer vision," *Journal of Outdoor Recreation and Tourism*, vol. 35, p. 100387, 2021, doi: [10.1016/j.jort.2021.100387](https://doi.org/10.1016/j.jort.2021.100387).
- [20] C. Sessions, S. A. Wood, S. Rabotyagov, and D. M. Fisher, "Measuring recreational visitation at U.S. National Parks with crowd-sourced photographs," *Journal of Environmental Management*, vol. 183, pp. 703–711, 2016, doi: [10.1016/j.jenvman.2016.09.018](https://doi.org/10.1016/j.jenvman.2016.09.018).